



larly, pattern feature recognition is another aspect intended for special components to recognise irregular placements and ensure accuracy.

Experts also try to find the best acceleration. In many cases, the acceleration of large components is set manually. Recent upgraded machines can automatically detect and accelerate to ensure the quality of placement and maximise the utilisation of equipment.

The next idea catching up is of PCB placement inspection. Before valuable components, such as BGAs and land-grid array (LGAs), are placed, interfering object detection is carried out in the placement area below these components. The foreign material/parts in excess are found in time to avoid placement defects. This can check whether there are any missing components under the shields before they are placed, which may play a good protective role in improving product quality.

The automotive electronics industry is expected to increase the demand for SMT continuously. Considering that automotive products have higher requirements for personal safety, service life, and outdoor use conditions, the reliability and durability needs of automotive electronic products are higher than those of others.

New PnP machines feature technologies like auto-cut sensor and advanced sensor technologies. Auto-cut sensor technology is a human safety feature that temporarily stops the process to avoid serious hazards or machine damage on human intervention. Advanced sensor technology helps a specific set of



Index PnP (Credit: <https://github.com/index-machines/index>)

sensors, like in-sensor, out-sensor, and waiting sensor, in tracking the PCB at every stage in the production process.

Auto-optimisation is another feature to consider as it can suggest feeder and nozzle measurements. There is also a feature called regular automatic detection for nozzles. The closed-loop control of the nozzle ensures that the nozzles on the placement head are always in a normal clean condition, and the contaminated and damaged nozzles are automatically replaced in time.

Choosing the best

The latest generation of the placement machines sets new records in speed, floorspace performance, and accuracy for high-volume production applications. Smart factory solutions are fast becoming a trend in the electronics manufacturing industry to cut down on costs and improve efficiency of production.

However, buyers should pay attention to performance and usability in relevance to their manufacturing needs. Machines are available in various price ranges with a wide range of customer support and

maintenance. Keeping in mind the investment on machine is vital to avoid spending a fortune on costly outsourced machines.

While the classic PnP vision systems are difficult to work in areas requiring production tolerances and deviations, Schubert Group—a company developing modular packaging machines—suggests its neuronal networks based on artificial intelligence could be a solution of the future. “The image processing system therefore learns on its own and sorts the products into classes independently in a longer learning process. This means that the parameters are selected by the system and no longer by the programmer. For customers, this means a significant step towards independence—In the future, they could actually use neuronal networks, under certain conditions, to teach their packaging machine to handle a new product on its own,” writes Dr Abdelmalek Nasraoui on the company website. **EFY**

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